

Shared evidence | Route and pump records

Double circulation: engineer the route

Name _____ Date _____

The route is a simplified adult circulation map. Pump numbers are constructed for learning.

Journey segment	Major vessel
Body → right atrium	Vena cava
Right ventricle → lungs	Pulmonary artery
Lungs → left atrium	Pulmonary veins
Left ventricle → body	Aorta

Pump record	Heart rate (beats/min)	Stroke volume (mL/beat)
P	80	55
Q	100	70

Cardiac output = heart rate × stroke volume. 1 L = 1000 mL. Atria receive blood; ventricles pump blood out. The two sides are separated by the septum.

W1. Trace the route. W2. Calculate both outputs. W3. Repair: “The thicker left ventricle proves it always pumps more blood than the right.”

Supported route | Follow the blood

Use the vessel table and calculator if needed.

S1. Complete a full route from body back to body, including lungs and the four chambers.

S2. Explain why the pulmonary artery is an artery even though its blood is relatively oxygen-poor.

S3. Calculate output for P and Q. Give one result in both mL/min and L/min.

S4. Explain one valve action and why the left-ventricle wall is thicker.

Standard route | Route-and-pump explanation

Use scientific names, mechanisms and units.

**M1. Trace body → lungs → body through the named vessels and chambers.
Explain the septum's role.**

**M2. Explain how chamber-wall thickness and valves support the circulation,
separating pressure from output.**

M3. Calculate both constructed outputs and their difference, with units.

**M4. Evaluate: "The faster pulse alone proves more blood is pumped each
minute." State what else is needed.**

Stretch route | Test the pump interpretation

Optional transfer. A constructed record R has 120 beats/min and 35 mL/beat. A record S has 70 beats/min and 70 mL/beat.

T1. Calculate both outputs. Which is greater? Explain why heart rate alone misleads.

T2. A faulty valve permits some backflow. Why might total chamber movement overestimate useful forward output?

T3. Explain why long-term unequal left/right outputs would be a problem for a linked circulation.

T4. A technician reports “5400 beats/min output”. Correct the quantity/unit error without diagnosing a person.

Exit and reflection | A direction and a quantity

Keep pressure, volume and time distinct.

E1. Which chamber pumps to the lungs?

E2. Explain the thicker left-ventricle wall using pressure.

E3. $60 \text{ mL/beat} \times 90 \text{ beats/min}$: show output and units.

L1. What did your audience ask? Improve one route or calculation label.

Next useful step

- A. Define vessels by direction.
- B. Trace all four chambers.
- C. Explain pressure and valves.
- D. Calculate output with units.